# Daily Problem: 11–Feb–2026

## Problem Statement

Let $R, S, T \subseteq \Sigma^*$ be languages (not necessarily regular). Show that:

(i) $\{\varepsilon\} \ \cup \ (R \cdot R^*) \ = \ R^*$.

(ii) $R \cdot R^* \ = \ R^* \cdot R$.

(iii) If $X = (R \cdot X) \ \cup \ S$, then $X = R^* \cdot S$.

If $Y = (R \cdot Y) \ \cup \ (S \cdot Y) \ \cup \ T$, what does it imply about $Y$?

## Solution

We use the standard definitions:

$$R^* = \bigcup_{n \geq 0} R^n, \qquad R^0 = \{\varepsilon\}, \qquad R^{n+1} = R \cdot R^n.$$

### (i) $\{\varepsilon\} \cup R \cdot R^* = R^*$

**Proof.** First note that by definition

$$R^* = R^0 \cup \bigcup_{n \geq 1} R^n = \{\varepsilon\} \ \cup \ \bigcup_{n \geq 1} R^n.$$

Also,

$$R \cdot R^* = R \cdot \left( \bigcup_{n \geq 0} R^n \right) = \bigcup_{n \geq 0} (R \cdot R^n) = \bigcup_{n \geq 0} R^{n+1} = \bigcup_{n \geq 1} R^n.$$

Therefore,

$$\{\varepsilon\} \cup (R \cdot R^*) = \{\varepsilon\} \cup \bigcup_{n \geq 1} R^n = \bigcup_{n \geq 0} R^n = R^*.$$

□

## (ii) $R \cdot R^* = R^* \cdot R$

**Proof.** Using the same expansion,

$$R \cdot R^* = \bigcup_{n \geq 0} R^{n+1} = \bigcup_{n \geq 1} R^n.$$

On the other hand,

$$R^* \cdot R = \left( \bigcup_{n \geq 0} R^n \right) \cdot R = \bigcup_{n \geq 0} (R^n \cdot R) = \bigcup_{n \geq 0} R^{n+1} = \bigcup_{n \geq 1} R^n.$$

Hence both are equal. □

## (iii) If $X = (R \cdot X) \cup S$, then $X = R^* \cdot S$

**Proof (two inclusions).**

**Step 1: Show $R^* \cdot S \subseteq X$.**

We prove by induction on $n \geq 0$ that

$$R^n \cdot S \subseteq X.$$

For $n = 0$, $R^0 \cdot S = \{\varepsilon\} \cdot S = S \subseteq X$ because $X = (R \cdot X) \cup S$. Assume $R^n \cdot S \subseteq X$. Then

$$R^{n+1} \cdot S = R \cdot (R^n \cdot S) \subseteq R \cdot X \subseteq X,$$

since $R^n \cdot S \subseteq X$ and $R \cdot X \subseteq X$ from $X = (R \cdot X) \cup S$. Thus the claim holds for all $n$, and taking the union over $n$ gives

$$R^* \cdot S = \bigcup_{n \geq 0} R^n \cdot S \subseteq X.$$

**Step 2: Show $X \subseteq R^* \cdot S$.**

Let $w \in X$. If $w \in S$, then $w \in R^0 \cdot S \subseteq R^* \cdot S$. Otherwise $w \notin S$. Since $X = (R \cdot X) \cup S$, we must have $w \in R \cdot X$, so $w = uv$ with $u \in R$ and $v \in X$.

If $v \in S$, then $w \in R \cdot S = R^1 \cdot S \subseteq R^* \cdot S$. If not, we repeat the same argument on $v$: either $v \in S$ or $v \in R \cdot X$, so we can peel off another prefix from $R$.

If this peeling process never reaches $S$, we would build an infinite decomposition

$$w = u_1 u_2 u_3 \cdots$$

with each $u_i \in R$, which is impossible because $w$ is a finite word. Hence after finitely many steps, say $n$ steps, we must reach a remainder in $S$. Thus $w \in R^n \cdot S \subseteq R^* \cdot S$.

Therefore $X \subseteq R^* \cdot S$.

Combining both inclusions, $X = R^* \cdot S$. □

# What does $Y = (R \cdot Y) \cup (S \cdot Y) \cup T$ imply about $Y$?

Let $U = R \cup S$. Then

$$(R \cdot Y) \cup (S \cdot Y) = (R \cup S) \cdot Y = U \cdot Y,$$

so the equation becomes

$$Y = (U \cdot Y) \cup T.$$

This is exactly the form in (iii), with $U$ playing the role of $R$ and $T$ playing the role of $S$. Therefore, by (iii),

$$Y = U^* \cdot T = (R \cup S)^* \cdot T.$$

# Final Conclusion

$$\boxed{Y = (R \cup S)^* \cdot T.}$$